

INTEGRATED MATH I

Chapter 1: Solving Linear Equations

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A. Warm-Up

1. $x + 5 = 12$

Hint: Subtract 5 from both sides.

Answer: 7

Solution:

$$x + 5 = 12 \rightarrow x = 7$$

2. $x + 2 = 13$

Hint: Subtract 2 from both sides.

Answer: 11

Solution:

$$x + 2 = 13 \rightarrow x = 11$$

3. $7x = 42$

Hint: Divide both sides by 7.

Answer: 6

Solution:

$$7x = 42 \rightarrow x = 6$$

B. Level Up

4. $2x + 3 = 17$

Hint: Isolate the term with the variable, then divide.

Answer: 7

Solution:

$$2x + 3 = 17 \rightarrow 2x = 14 \rightarrow x = 7$$

5. $2(x + 1) = 12$

Hint: Divide both sides by 2, then subtract 1.

Answer: 5

Solution:

$$2(x + 1) = 12 \rightarrow x + 1 = 6 \rightarrow x = 5$$

6. $4x + 8 = 24$

Hint: Isolate the term with the variable, then divide.

Answer: 4

Solution:

$$4x + 8 = 24 \rightarrow 4x = 16 \rightarrow x = 4$$

C. Spot the Mistake

7. A student solved: $2x + 4 = 16$ $2x = 16$ $x = 16$ Where is the mistake?

Answer: 6

Solution:

Correct steps: $2x + 4 = 16 \rightarrow 2x = 12 \rightarrow x = 6$

D. Choose the Correct Move

(No items generated)

E. Match / Graph / Short Tasks

(No items generated)

F. Challenge

8. Solve: $4(x + 5) - 2 = 30$

Hint: Undo operations in reverse order (add/sub first, then multiply/divide).

Answer: 3

Solution:

Solve: $4(x + 5) - 2 = 30 \rightarrow 4(x + 5) = 30 + 2 \rightarrow x + 5 = 8 \rightarrow x = 3$